

Bianca Bai

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EDUCATION

University of Waterloo

Bachelor of Software Engineering, Honours

Sep 2024 – Apr 2029

4.0/4.0 GPA

EXPERIENCE

Machine Learning Engineer

Huawei Technologies Canada Co., Ltd.

May 2025 – Aug 2025

Ottawa, Canada

- Engineered **CNN** architectures using **PyTorch** to precisely calibrate S21 filters, significantly improving signal integrity for optical transceiver systems.
- Championed project selection over **5** competing approaches by headquarters, where it was subsequently integrated into the production system.
- Simulated an optical transceiver pipeline using **Numpy** and **Scipy** to generate high-fidelity training data.
- Boosted model accuracy by elevating SNR from **20 dB to 40 dB** through targeted optimization of hyperparameters and custom loss function.
- Reduced training time by over **50%** through dynamic optimization of learning rates utilizing LR range test.
- Accommodated over **5 types** of optical channel impairments by designing **6** model variations.
- Utilized **Matplotlib** to visualize convolutional layer weights across time and frequency domains, yielding critical insights to adjust channel imbalance and improve model stability.

Software Development Intern

Xintongda SCIENCE & TECHNOLOGY Group Co.

Apr 2024 – Aug 2024

- Deployed and configured a robust **CoreDNS** service on **Linux** for the company intranet, ensuring high-availability name resolution for **20+** internal users and critical services
- Created an application using **Go** and **REST** API to automate GitLab runner registration, reducing CI pipeline setup time by **50%**.
- Developed a componentized frontend for the company's database management application using **Vue.js**, increasing development efficiency by **30%**.

PROJECTS

Moodii |

Apr 2024 – Aug 2024

- Developed an AI-powered Android app in **Kotlin** with Jetpack Compose, enabling users to log moods and track emotion trends via voice recordings.
- Designed and implemented a **Spring Boot** backend with **15+** RESTful APIs, enabling secure user authentication, efficient data storage, and advanced emotion trend analytics.
- Integrated a **CNN+LSTM** Speech Emotion Recognition model through **Flask**, achieving **81%** accuracy.
- Orchestrated the containerization of backend and ML services using **Docker**, deploying to **Azure** with automated CI/CD pipelines to ensure rapid, reliable deployments and horizontal scalability

EyeSee |

Nov 2024 – Jan 2025

- Developed an IOS application that detects and reads texts on street view text (SVT) images using **React Native Expo**, **Node.js**, and **Typescript**.
- Constructed and trained a CRNN-based OCR model using **TensorFlow**; fine-tuned the model to achieve an accuracy of **84%** for SVT images.
- Designed and implemented advanced data preprocessing pipelines using **NumPy** and **OpenCV**, cleaning and augmenting large, noisy datasets of SVT images to significantly enhance model training efficiency and performance.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Kotlin, SQL, Go, HTML & CSS, Typescript & JavaScript

Frameworks and Libraries: PyTorch, TensorFlow, Spring Boot, Node.js, React.js, Django, NumPy

Developer Tools: AWS, Azure, Docker, MongoDB, MySQL, Git, GitHub, Linux, Postman, Docker